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/**
 * Enterprise Sales Dashboard - Google Apps Script Backend
 * Method: Spreadsheet-Bound Web App Script Framework
 */

function doGet() {
  return HtmlService.createTemplateFromFile('Index')
    .evaluate()
    .setTitle('Sales Performance Dashboard')
    .addMetaTag('viewport', 'width=device-width, initial-scale=1')
    .setXFrameOptionsMode(HtmlService.XFrameOptionsMode.ALLOWALL);
}

/**
 * Server-side include function to inline HTML components (CSS and JavaScript)
 */
function include(filename) {
  return HtmlService.createHtmlOutputFromFile(filename).getContent();
}

/**
 * Fetches, cleans, and normalizes spreadsheet transactions for the dashboard
 */
function getDashboardData() {
  try {
    var sheet = SpreadsheetApp.getActiveSpreadsheet().getActiveSheet();
    var data = sheet.getDataRange().getValues();

    if (data.length <= 1) {
      throw new Error("The active sheet contains no data rows. Please ensure your sales data is imported into the current sheet.");
    }
  }
}

```

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}
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```
var headers = data[0];
```

```
// Map column names dynamically to guarantee resilience against shifted positions
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```
var idxId = headers.indexOf("Transaction_ID");
```

```
var idxDate = headers.indexOf("Date");
```

```
var idxRegion = headers.indexOf("Region");
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```
var idxCat = headers.indexOf("Product_Category");
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```
var idxSku = headers.indexOf("Product_SKU");
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```
var idxQty = headers.indexOf("Quantity");
```

```
var idxPrice = headers.indexOf("Unit_Price");
```

```
var idxDiscount = headers.indexOf("Discount_Applied");
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```
var idxRating = headers.indexOf("Customer_Rating");
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```
var idxMember = headers.indexOf("Is_Member");
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```
var idxSales = headers.indexOf("Total_Sales");
```

```
// Structural validation constraint check
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```
if (idxId === -1 || idxDate === -1 || idxRegion === -1 || idxCat === -1 ||
```

```
    idxQty === -1 || idxPrice === -1 || idxDiscount === -1 ||
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```
    idxRating === -1 || idxMember === -1 || idxSales === -1) {
```

```
    throw new Error("Missing critical column headers. Please verify that your spreadsheet  
contains exactly the headers present in your CSV sample (e.g., Transaction_ID, Date, Region,  
Product_Category, etc.).");
```

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}
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```
var normalizedDataset = [];
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```
for (var i = 1; i < data.length; i++) {
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```
    var row = data[i];
```

```
// Filter out trailing blank lines or partial system artifacts
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if (row.length === 0 || row.every(function(cell) { return cell === ""; })) continue;
if (row[idxId] === "" || row[idxId] === null || row[idxId] === undefined) continue;

// 1. Server-Side Normalization: Dates
var rawDate = row[idxDate];
var formattedIsoDate = "";
var parsedYear = 2023;
var parsedMonthIndex = 0; // 0-based for JS arrays

if (rawDate instanceof Date) {
    parsedYear = rawDate.getFullYear();
    parsedMonthIndex = rawDate.getMonth();

    formattedIsoDate = parsedYear + "-" + ("0" + (parsedMonthIndex + 1)).slice(-2) + "-" + ("0" +
rawDate.getDate()).slice(-2);
} else if (rawDate && typeof rawDate === "string") {
    var dateObj = new Date(rawDate.trim());
    if (!isNaN(dateObj.getTime())) {
        parsedYear = dateObj.getFullYear();
        parsedMonthIndex = dateObj.getMonth();

        formattedIsoDate = parsedYear + "-" + ("0" + (parsedMonthIndex + 1)).slice(-2) + "-" + ("0" +
dateObj.getDate()).slice(-2);
    }
}

// 2. Server-Side Normalization: Numbers (Removes formatting symbols like '$' or commas)
var qty = helperCleanNumber(row[idxQty]);
var price = helperCleanNumber(row[idxPrice]);
var discount = helperCleanNumber(row[idxDiscount]);
var rating = helperCleanNumber(row[idxRating]);
var sales = helperCleanNumber(row[idxSales]);

// 3. Server-Side Normalization: Booleans

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var rawMember = String(row[idxMember]).trim().toLowerCase();
var isMember = (rawMember === "yes" || rawMember === "true" || rawMember === "1");

normalizedDataset.push({
  id: row[idxId],
  date: formattedIsoDate,
  year: parsedYear,
  month: parsedMonthIndex,
  region: String(row[idxRegion]).trim(),
  category: String(row[idxCat]).trim(),
  sku: idxSku !== -1 ? String(row[idxSku]).trim() : "N/A",
  quantity: qty,
  unitPrice: price,
  discount: discount,
  rating: rating,
  isMember: isMember,
  totalSales: sales
});
}

return normalizedDataset;
} catch (error) {
  throw new Error("Execution Framework Error: " + error.message);
}
}

/**
 * Sanitizes and extracts numerical components cleanly
 */
function helperCleanNumber(val) {
  if (val === "" || val === null || val === undefined) return 0;

```

```
if (typeof val === "number") return val;
var sanitizedStr = String(val).replace(/[\$,\s]/g, "");
var conversion = Number(sanitizedStr);
return isNaN(conversion) ? 0 : conversion;
}
```